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ES 668 Assignment 4.2

1. Numerical methods to solve PDEs

Answer:

Finite difference methods: Forward difference, Backward difference, Central difference

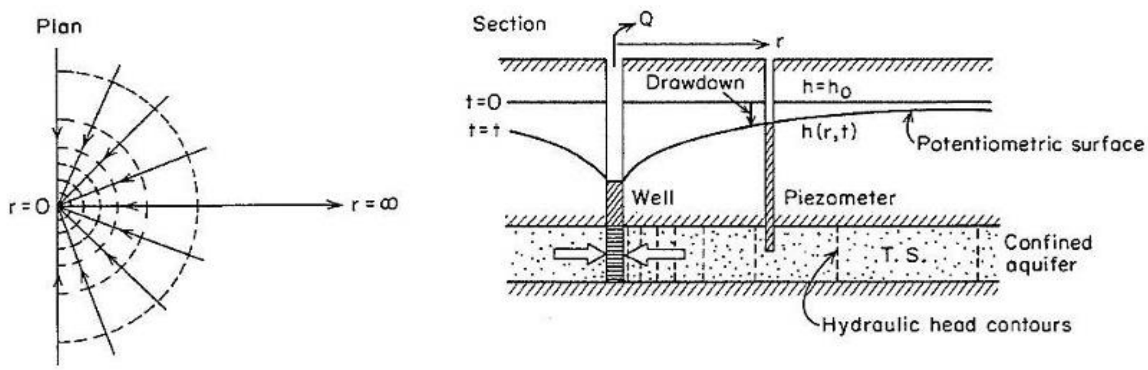
Finite element methods: Galerkin method, Ritz method, Variational method

Analytical methods: Separation of variables, Fourier series, Laplace transform

EXERCISE PROBLEM

2. The partial differential equation given below describes the saturated flow in radial coordinates in a confined aquifer with transmissivity 'T' (4639 m^2/day) and storativity 'S' (4.493×10^{-4}). The mathematical region of flow, as given in the plan view is a horizontal 1-D line through the aquifer, from r = 0 at the well to r = ∞ at the infinite extremity.

$$\left(\frac{S}{T}\right) \cdot \frac{\partial h}{\partial t} = \frac{1}{r} \cdot \frac{\partial}{\partial r}\left(r \frac{\partial h}{\partial r}\right)$$



- The initial condition is $h(r, 0) = h_0$ for all r, where h_0 is the constant initial hydraulic head (1000 m).
- The left boundary (r→1 m) at the well assumes a constant pumping rate Q (8640 m^3/day), such that:

$$\lim_{r \to 0}\left(r \frac{\partial h}{\partial r}\right) = \frac{Q}{2\pi T} \qquad \text{for } t > 0$$

- The right boundary condition assumes no drawdown at the infinite boundary: $h(\infty, t) = h_0$ for all t. This can be modified to a zero gradient for obtaining the numerical solution at a large distance (100 km).

(a) Solve the PDE in MATLAB (maximum radial distance of 100 km, maximum time of 100 days) with 'pdepe' and plot the solution h(r, t) as a surface plot.

(b) Plot the drawdown $\{h_0 - h(r, t)\}$ vs time at a radial distance of 100 m from the pumping well.

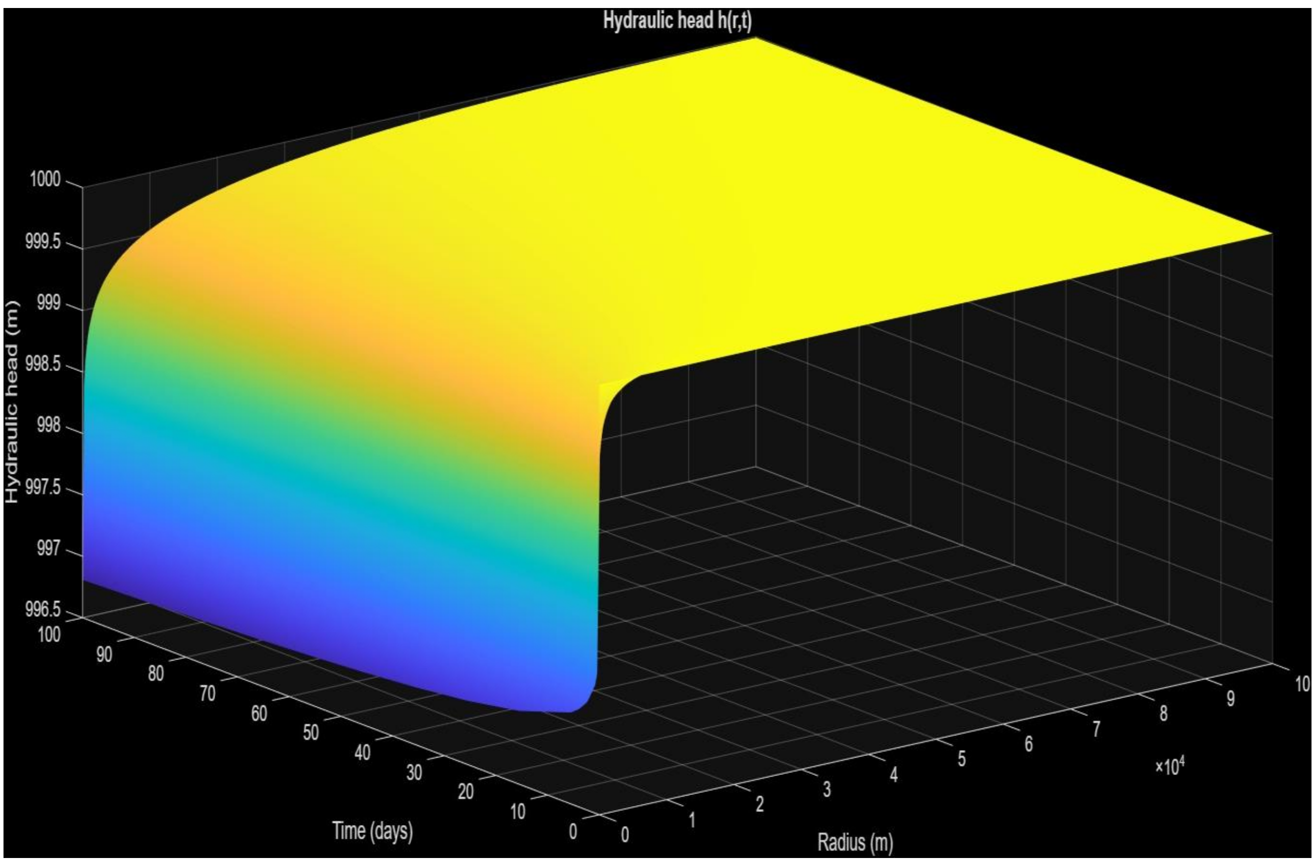
(c) Plot the drawdown vs radial distance at any appropriate time.

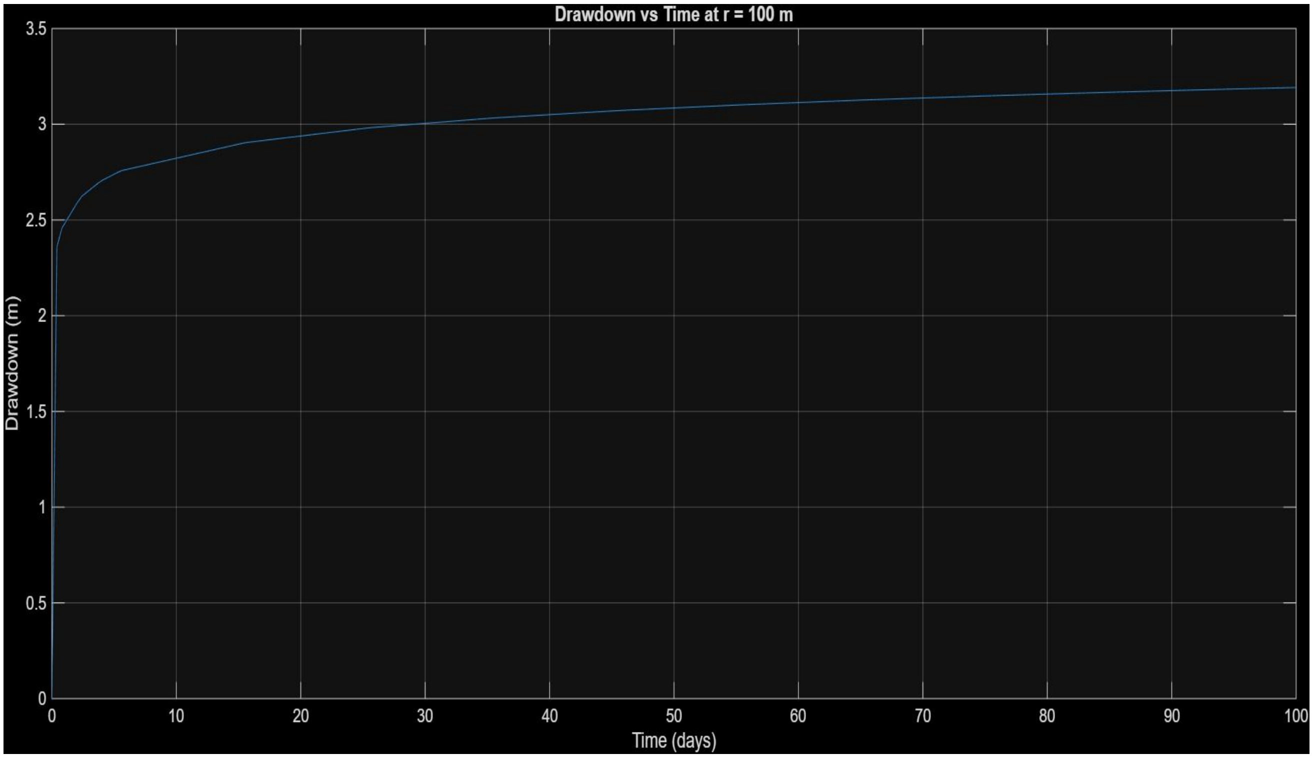
Answer: Script- `function Aryan_25M1429_Assignment_4_2`

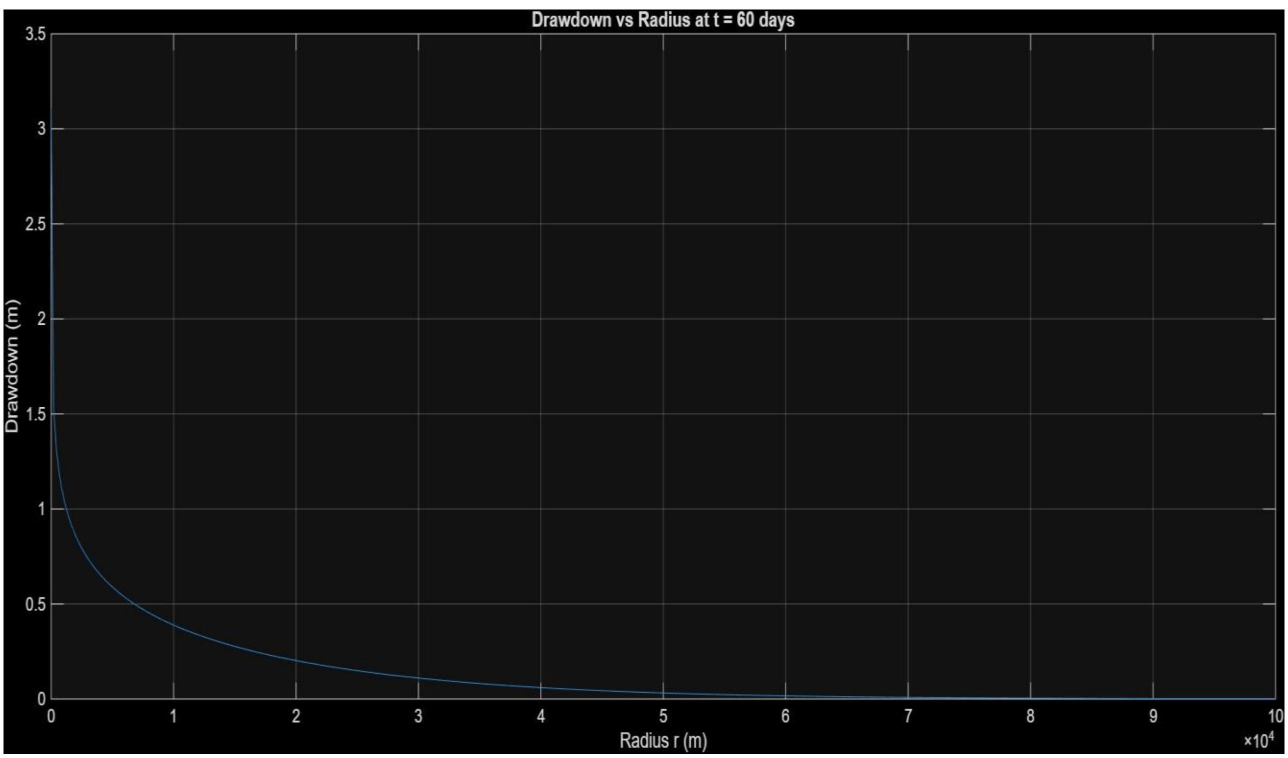
```matlab
% Parameters
T = 4639; % transmissivity (m^2/day)
S = 0.0004493; % storativity
Q = 8640; % pumping rate (m^3/day)
h0 = 1000; % initial head (m)
rmax = 100000; % 100 km
tmax = 100; % 100 days
% time and distance span
r = linspace(1, rmax, 500);
t = linspace(0, tmax, 250);
% Solve PDE
sol = pdepe(1,@pbpde,@icpb,@bcpb,r,t);
h = sol(:,:,1);
% (a) Hydraulic head over space and time
figure;
surf(r,t,h);
xlabel('Radius (m)');
ylabel('Time (days)');
zlabel('Hydraulic head (m)');
title('Hydraulic head h(r,t)');
shading interp
grid on
% (b) Drawdown vs time at r = 100 m
r_obs = 100;
[~, idx] = min(abs(r - r_obs));
drawdown_t = h0 - h(:,idx);
figure;
plot(t,drawdown_t);
xlabel('Time (days)');
ylabel('Drawdown (m)');
title('Drawdown vs Time at r = 100 m');
grid on
% (c) Drawdown vs radius at t = 60 days
t_obs = 60;
[~, idt] = min(abs(t - t_obs));
drawdown_r = h0 - h(idt,:);
figure;
plot(r,drawdown_r);
xlabel('Radius r (m)');
ylabel('Drawdown (m)');
title('Drawdown vs Radius at t = 60 days');
grid on
% defined functions
function [c,f,s] = pbpde(r,t,h,dhdr)
c = S/T;
f = dhdr;
s = 0;
```

```
end
% initial condition
function hini = icpb(r)
hini = h0;
end
% boundary condition
function [pl,ql,pr,qr] = bcpb(rl,hl,rr,dhdr,t)
pl = -Q/(2*pi*T);
ql = 1;
pr = 0;
qr = 1;
end
end
```

Graph-




Drawdown vs Time at r = 100 m
Drawdown (m)
0
0.5
1
1.5
2
2.5
3
3.5
0
10
20
30
40
50
60
70
80
90
100
Time (days)


Drawdown vs Radius at t = 60 days
Drawdown (m)
0
0.5
1
1.5
2
2.5
3
3.5
0
1
2
3
4
5
6
7
8
9
10
×10^4
Radius r (m)